

Big Data and Parallel Programming

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Project Introduction

Project Overview

- At the end of the module your team (with 3 or 4 members) will have to deliver:
 - An implemented solution for dealing with a supervised distributed machine learning task.
 - A compact presentation explaining the methodology you used, the results and their analysis.
 - The developed codes and datasets used for the class project paper.
 - Filled forms regarding the evaluation of the individual members softskills.
- The main aspects of the project are to evaluate the impact of using spark in datasets with different sizes and configurations to understand when is it worth to use it in **classification** (a.k.a supervised machine learning) problems.

Class Project Roadmap

1st Step: Database Selection

2nd Step: Elaborate a short text about the chosen datasets.

- What is the problem?
- How many classes?
- How many attributes?
- What type of attributes?
- Relevant Additional information

Class Project Roadmap

3rd Step: Think about your big data parallelization strategy. How are you going to achieve it?

4th Step: Define your evaluation plan. I.e. How are you going to know whether or not the parallelization was better in comparison with a baseline approach? What is it going to be your parallelization approach? What is your definition of better? (Faster? More Reliable? Other ways?)

Class Project Roadmap

5th Step: Execute and evaluate your experiments.

6th Step: Elaborate the project presentation and submit it.

Parallel Steps: Fill out the forms related to soft skills and teamwork in the project.

Project Requirements

- Your team is expected to at least:
 - Perform experiments using four datasets (two small/medium and two large).
 - Execute and Evaluate the performance of a baseline approach without parallelization.
 - Execute and Evaluate the performance of a big data parallel approach with different parameterizations (at least three different configurations) with at least 2 different classifiers.
 - Perform an in-depth analysis of the results based on your proposed evaluation plan.
 - Fill out the forms for the soft skills assessment of the team members during the project development.
 - Prepare a compact presentation for the project and code interview.

Project Evaluation

For evaluating the class project, we are going to have **in-person interviews** with the teams, where all members of the teams must show that they are able to understand and modify everything about the code and the project. (i.e. discussing the experimental and methodological choices and discuss the results).

During these interviews, we will ask you **individually** to do at least one of these things:

- Explain how something specific works,
- Change things in the code,
- Add things to the code,
- Remove things (with/without breaking) the code
- Fix small errors we might have introduced in your code.
- Discuss methodological and experimental parts of the project and the obtained results.

During the interview you will not be allowed to consult other sources of information or your colleagues. Therefore it is really important that you own the whole code, answers such as **“I did not do that part”** will result in a **failing grade** regarding the project evaluation.

Also it is important to highlight that the code must be working properly, and the experiments were properly designed and that all the project requirements have been met for the student(s) to obtain a passing grade.

Deadlines - 1st Evaluation Opportunity

Project Submission: 25th of May of 2026

In-Person Project Interviews/Evaluations: 27-28th of May of 2026

Deadlines - 2nd Evaluation Opportunity

Project Submission: 31th of August of 2026

In-Person Project Interviews/Evaluations: To be disclosed but between the day after the submission deadline and the 18th of September 2026.

Deadlines - 3rd Evaluation Opportunity

Project Submission: 19th of November of 2026

In-Person Project Interviews/Evaluations: To be disclosed but between the day after the submission deadline and the 5th of December.

Today's Plan

1st: We need to double check everyone is in a team in blackboard.

2nd: You will need to investigate datasets that you find interesting relevant. The idea is that you choose at least two datasets, two being small/medium and another two which needs to be a big dataset.

- You can search for datasets in online resources like kaggle.com or the UCI datasets.
 - **Attention:**
 - Each specific dataset can only be used by one team.
 - We will coordinate dataset usage rights using Blackboard forum. First team to post that they will use a certain dataset is the one who will be allowed to use it.

3rd: We must ensure everyone filled out the forms for the soft skills assessment.

4rd: We need to figure out the time slots for our **in-person** groups supervision.